A Lehmann–Romano Sufficiency Theorem for the Normal Orthant Problem

1 Why exact size at a finite null point is needed

Let

$$X \sim N(\mu, \Sigma), \qquad \Sigma > 0,$$

where Σ is known, and consider

$$H_0 : \mu \in \Lambda_0 = \{\mu \in \mathbf{R}^k : \mu \leq 0\} \qquad \text{versus} \qquad H_1 : \mu \in \Lambda_1 = \mathbf{R}^k \setminus \Lambda_0.$$

A level-α test satisfies

$$\sup_{\mu \in \Lambda_0} E_\mu[\phi(X)] \leq \alpha.$$

Closed convexity of the acceptance region cannot by itself imply α-admissibility unless the test actually uses the available size. For example, the zero test $\phi \equiv 0$ has acceptance region $A_\phi = \mathbf{R}^k$, which is closed and convex, and it satisfies the size constraint for every $\alpha > 0$. But it is dominated by any nontrivial level-α test with positive power on Λ_1. Thus a theorem of the form "closed convex acceptance region implies admissibility" must rule out such tests.

The exact-size condition also appears directly in the Lehmann–Romano proof. Their convex-acceptance-region theorem implies that if a competing test has power at least as large as ϕ_0 at every alternative, then the competing test must reject at least wherever ϕ_0 rejects, up to null sets. To turn this inclusion into α-admissibility, one needs a null point at which ϕ_0 already has rejection probability α. Otherwise, a strict enlargement of the rejection region may still remain below level α.

Therefore the natural statement is not the unrestricted equivalence as written, but the following sufficiency result: if the acceptance region is closed and convex, and if the test has size α at some finite null point, then the test is α-admissible.

2 The theorem and proof

We use the following result from Lehmann and Romano, *Testing Statistical Hypotheses*, third edition, Section 6.7, Theorem 6.7.1 and Corollary 6.7.1.

Lehmann–Romano result used. Consider an exponential family with density

$$p_\theta(x) = C(\theta) \exp\left\{ \sum_{j=1}^{s} \theta_j T_j(x) \right\}$$

with natural parameter space Ω. Let Ω_H and Ω' be disjoint nonempty subsets of Ω. Let ϕ_0 be a nonrandomized test based on $T = (T_1, \ldots, T_s)$, and suppose its acceptance region $A_0 \subseteq \mathbf{R}^s$ is closed and convex. Assume that whenever

$$A_0 \cap \left\{ t : \sum_{j=1}^{s} a_j t_j > c \right\} = \emptyset,$$

there exist $\theta^* \in \Omega$ and a sequence $\lambda_n \to \infty$ such that

$$\theta^* + \lambda_n a \in \Omega'.$$

Then any other acceptance region A satisfying

$$P_\theta(T \in A) \le P_\theta(T \in A_0) \qquad \text{for all } \theta \in \Omega'$$

is contained in A_0, except for a null set. Consequently, the test with acceptance region A_0 is d-admissible. If, in addition, it has size α and there is a finite point $\theta_0 \in \overline{\Omega_H}$ such that

$$E_{\theta_0}[\phi_0(X)] = \alpha,$$

then it is α-admissible. The theorem also extends to randomized competitors.

Theorem. Let $X \sim N(\mu, \Sigma)$, where Σ is known and positive definite and $\mu \in \mathbf{R}^k$. Consider testing

$$H_0 : \mu \in \Lambda_0 = \{\mu \in \mathbf{R}^k : \mu \le 0\} \qquad \text{versus} \qquad H_1 : \mu \in \Lambda_1 = \mathbf{R}^k \setminus \Lambda_0.$$

Let $0 < \alpha < 1$. Let ϕ_0 be a nonrandomized level-α test with acceptance region

$$A_0 = \{x \in \mathbf{R}^k : \phi_0(x) = 0\}.$$

Assume that A_0 is closed and convex, and that there exists a finite point $\mu_0 \in \Lambda_0$ such that

$$E_{\mu_0}[\phi_0(X)] = \alpha.$$

Then ϕ_0 is α-admissible for testing H_0 against H_1.

Proof. Write the normal location family in canonical exponential-family form. Put

$$\theta = \Sigma^{-1}\mu, \qquad \text{so that} \qquad \mu = \Sigma\theta.$$

With respect to the measure with density proportional to $\exp\{-x'\Sigma^{-1}x/2\}$, the density can be written as

$$p_\theta(x) = C(\theta) \exp\{\theta' x\},$$

where

$$C(\theta) = \exp\left\{ -\frac{1}{2}\theta'\Sigma\theta \right\}.$$

Thus the sufficient statistic is

$$T(X) = X,$$

the natural parameter space is

$$\Omega = \mathbf{R}^k,$$

and the null and alternative parameter sets in natural coordinates are

$$\Omega_H = \{\theta \in \mathbf{R}^k : \Sigma\theta \le 0\}, \qquad \Omega_K = \{\theta \in \mathbf{R}^k : \Sigma\theta \not\le 0\}.$$

We verify the ray condition in the Lehmann–Romano theorem. Suppose

$$A_0 \cap \{x : a'x > c\} = \emptyset$$

for some $a \in \mathbf{R}^k$ and $c \in \mathbf{R}$. If $a = 0$, the condition is trivial, since one may choose any $\theta^* \in \Omega_K$. Hence assume $a \neq 0$. The displayed condition says

$$A_0 \subseteq \{x : a'x \le c\}.$$

We claim that

$$\Sigma a \not\le 0.$$

Suppose, toward a contradiction, that $\Sigma a \le 0$. Then, for every $\lambda > 0$,

$$\theta_\lambda = \lambda a \in \Omega_H,$$

because its mean parameter is

$$\mu_\lambda = \Sigma\theta_\lambda = \lambda\Sigma a \le 0.$$

Under θ_λ, equivalently under $X \sim N(\lambda\Sigma a, \Sigma)$,

$$a'X \sim N\left(\lambda a'\Sigma a,\, a'\Sigma a\right).$$

Since Σ is positive definite and $a \neq 0$,

$$a'\Sigma a > 0.$$

Therefore

$$P_{\theta_\lambda}(a'X > c) \to 1 \qquad \text{as } \lambda \to \infty.$$

Because $A_0 \subseteq \{x : a'x \le c\}$, the rejection region A_0^c contains $\{x : a'x > c\}$. Hence

$$E_{\theta_\lambda}[\phi_0(X)] = P_{\theta_\lambda}(X \in A_0^c) \ge P_{\theta_\lambda}(a'X > c) \to 1.$$

This contradicts the level-α condition, since $0 < \alpha < 1$ and all $\theta_\lambda \in \Omega_H$. Thus $\Sigma a \not\le 0$.

It follows that, for every $\lambda > 0$,

$$\Sigma(\lambda a) = \lambda\Sigma a$$

has at least one positive coordinate. Therefore

$$\lambda a \in \Omega_K \qquad \text{for every } \lambda > 0.$$

Taking $\theta^* = 0$ and any sequence $\lambda_n \to \infty$, the Lehmann–Romano ray condition is satisfied.

The hypotheses of the Lehmann–Romano theorem are therefore satisfied with $\Omega' = \Omega_K$. Since $\mu_0 \in \Lambda_0$, the corresponding natural parameter

$$\theta_0 = \Sigma^{-1}\mu_0$$

belongs to Ω_H, and by assumption

$$E_{\theta_0}[\phi_0(X)] = \alpha.$$

The Lehmann–Romano corollary therefore implies that ϕ_0 is α-admissible.

For completeness, here is the final implication explicitly. Let ϕ be any level-α test such that

$$E_\mu[\phi(X)] \geq E_\mu[\phi_0(X)] \qquad \text{for all } \mu \in \Lambda_1.$$

Equivalently,

$$P_\mu(\text{accept under } \phi) \leq P_\mu(X \in A_0) \qquad \text{for all } \mu \in \Lambda_1.$$

By the Lehmann–Romano theorem, $\phi \geq \phi_0$ almost everywhere. If the inequality were strict on a set of positive Lebesgue measure, then, because every normal density is strictly positive on $\mathbf{R}^k$,

$$E_{\mu_0}[\phi(X)] > E_{\mu_0}[\phi_0(X)] = \alpha,$$

contradicting the level-α constraint. Hence $\phi = \phi_0$ almost everywhere, and therefore

$$E_\mu[\phi(X)] = E_\mu[\phi_0(X)] \qquad \text{for all } \mu \in \Lambda_1.$$

This is exactly α-admissibility. $\square$